

Final Project

Causal Inference

Nathan Acciai

July 2026

Contents

1	Introduction	2
2	Theoretical Framework and Instrumental Variables Motivation	3
2.1	From Assignment to Treatment: the role of non-compliance . . .	3
3	Non Compliance Problem	4
4	Variable Anlaysis	5
4.1	Analysis on full dataset	7
4.2	Subset Analysis by Assignment	7
4.3	Subset Analysis by Receipt Indicators	7
5	Outcome Analysis	8
6	Analysis of Unconfoundedness	8
6.1	Unconfoundedness of Assignment	8
6.2	Unconfoundedness of Treatment Receipt	8
6.3	The importance of Unconfoundedness	9
7	Intention-To-Treat Effect	9
8	Compliance Type	10
9	Monotonicity Assumption	11
10	Exclusion Restriction	11
10.1	Exclusion Restriction Never-takers	11
10.1.1	Ground suspicion	11
10.1.2	Plausibility Assessment	11
10.2	Exclusion Restriction Always-taker	12
11	Complier Average Causal Effect (CACE)	12

12 CACE vs ITT	13
13 Method of Moment	13
13.1 Theroetical Introduction	13
13.2 Estimation Results	15

1 Introduction

The objective in this work is to analyze the efficiency of social intervention, taking into account the actual behavior of the units in this case study. The dataset chosen is JobsII, was a randomized field experiment, which analyses the effectiveness of a vocational training and psychological support programme aimed at unemployed workers. On the one hand, the programme aims to promote employment; on the other, to reduce the negative impact of unemployment on mental health. A brief summary of the dataset composition:

- **Total Units:** 899
- **Covariates:**
 - **Age:** Measured in years
 - **Sex:** Indicate the sex, 1 = *female*
 - **Married:** Indicator married, 1 = *married*
 - **NonWhite:** Indicator of race, 1 = *nonwhite*
 - **Educ:** indicator of higher education, 1 = *collegeormore*
 - **Income:** variabel of income
 - **Depress_Pre:** Indicate the level of depression before treatment
- **Assignment variable:**
 - **Z:** Indicator variable for whether participant was randomly selected, 1 = assignment to participation
- **Treatment receipt indicator:**
 - **W:**Indicator variable for whether participant actually participated, 1 = participation
- **Outcome:**
 - **Job_seek:** A continuous scale measuring the level of self-efficacy in job-search with values of 1 to 5

2 Theoretical Framework and Instrumental Variables Motivation

To formalize the problem of non-compliance, we introduce the potential outcomes framework. For each unit i , let:

$$Y_i(1), Y_i(0)$$

denote the potential outcomes under treatment and no treatment, respectively. Similarly, we define the potential treatment status as:

$$W_i(1), W_i(0)$$

where $W_i(1)$ is the treatment received if assigned to treatment ($Z_i = 1$), and $W_i(0)$ if assigned to control ($Z_i = 0$).

The observed treatment and outcome can then be written as:

$$W_i = W_i(Z_i), \quad Y_i = Y_i(W_i(Z_i)).$$

2.1 From Assignment to Treatment: the role of non-compliance

In a perfect randomized experiment, we would have:

$$W_i = Z_i$$

so that assignment directly determines treatment receipt, however, in the presence of non-compliance this equality does not hold, and we observe:

$$W_i \neq Z_i \quad \text{for some units.}$$

This implies that the treatment status W_i is no longer randomly assigned, even though Z_i is.

Because individuals self-select into treatment participation, W_i is generally correlated with unobserved determinants of the outcome Y_i , such as motivation, psychological resilience, or external support networks. Formally:

$$\mathbb{E}[Y_i(0) \mid W_i = 1] \neq \mathbb{E}[Y_i(0) \mid W_i = 0],$$

which implies that a naive comparison between treated and untreated units does not identify a causal effect, this issue is known as **selection bias** or **endogeneity of treatment receipt**, although W_i is endogenous, the assignment variable Z_i is generated through randomization and satisfies:

$$Z_i \perp (Y_i(0), Y_i(1), W_i(0), W_i(1), X_i).$$

Moreover, Z_i affects the outcome Y_i only through its effect on W_i , satisfying the exclusion restriction, these two conditions allow us to use Z_i as an **instrumental variable** for W_i .

A valid instrument must satisfy:

- **Relevance:**

$$\mathbb{P}(W_i = 1 \mid Z_i = 1) \neq \mathbb{P}(W_i = 1 \mid Z_i = 0)$$

which is confirmed empirically by the compliance rate observed in the data.

- **Endogeneity:**

$$Z_i \perp (Y_i(0), Y_i(1))$$

guaranteed by random assignment.

- **Exclusion restriction:**

$$Z_i \rightarrow Y_i \text{ only through } W_i.$$

3 Non Compliance Problem

Non-compliance arises when the treatment actually received W_i does not coincide with the randomized assignment Z_i , i.e. $W_i \neq Z_i$ for at least some units. In the JOBS II experiment, Z_i represents the random assignment to the program (the invitation to participate), while W_i indicates actual participation, the difference between these two variables reflects behavioral responses of individuals after assignment, this discrepancy can be directly assessed in the data through the joint distribution of (Z, W) , which provides evidence on the extent and structure of non-compliance.

- **Z:** Represent the randomized assignment (the invitation to take part)
- **W:** Represent the effective treatment received (actual participation)

Clearly, there are two broad categories that distinguish the issue of non-compliance; these are distinguished by looking at how the values of Z and W are defined: The two categories are:

- **One-Side Noncompliance:** This occurs when access to the treatment is strictly limited to those assigned to the treatment group. Those assigned to the control group ($Z=0$) have no chance of receiving the treatment ($W=0$). In this case, non-compliance occurs only within the treatment group .
- **Two-side Noncompliance:** This occurs when both the subjects in the treatment group can refuse it and the subjects in the control group can obtain it from external sources .

Analysing this case of study we can conclude that we are dealing with a case of **One-Side Non Compliance**, because the units which have not been assigned cannot attend the course in person and register as participants. Another thing that confirms this hypothesis is the data, in particular the joint frequency matrix. Let us therefore explain the meaning of the figures shown in table:

Table 1: Joint frequency matrix between assignment (z) and treatment receipt (w)

	$w = 0$	$w = 1$
$z = 0$	299	0
$z = 1$	228	372

- **First Row** $z = 0$:

- $w = 0 = 299$ All those in the control group (not assigned to the programme) who are not taking part in the treatment. These figures are not a pure sample because this number includes people who would have taken part if they'd been assigned ($z=1$) but who are not taking part here because they were not invited.
- $w = 1: 0$ no one assigned to control group took part in the treatment. This number

- **Second row** $z = 1$:

- $w = 0: 228$ Represent the number of the unit's assigned at the program but don't taken part.
- $w = 1: 372$ Representt the unit's that are assigned at the program who have actually agreed to take part.

Rerturning to the definition of the noncompliance problem, the value of the intersection between $z = 0, w = 1$, which is 0, demonstrates precisely that the units in the control group did not have the opportunity to access the treatment; hence, we refer to this as **One-Side Non-Compliance Problem**

4 Variable Anlaysis

In this section we analyze the variable inside the dataset, as specified the variable "income", as it is categorical variable, has been trasformed into a dummy variable.

Table 2: Analysis of variable means

Variable	Full sample	Control (Z=0)	Treatment (Z=1)	W=0	W=1
age	37.5651	37.3173	37.6885	36.4040	39.2099
sex	0.5362	0.5753	0.5167	0.5769	0.4785
married	0.4538	0.4515	0.4550	0.4459	0.4651
nonwhite	0.1691	0.1739	0.1667	0.1822	0.1505
educ	0.6418	0.6388	0.6433	0.6205	0.6720
depress_pre	1.8700	1.7837	1.7203	1.7659	1.7066
z	0.6674	0.0000	1.0000	0.4326	1.0000
w	0.4138	0.0000	0.6200	0.0000	1.0000
job_seek	4.0433	3.9983	4.0658	3.9930	4.1146
incomelt15k	0.1824	0.1873	0.1800	0.2106	0.1425
income15t24k	0.2291	0.2207	0.2333	0.2277	0.2312
income25t39k	0.2425	0.2308	0.2483	0.2486	0.2339
income40t49k	0.1224	0.1438	0.1117	0.1139	0.1344
income50k+	0.2236	0.2174	0.2267	0.1992	0.2581

4.1 Analysis on full dataset

The JOBS II sample consists of unemployed individuals with an average age of 37.6 years. Women slightly predominate (53.6%), while 45.4% are married and 16.9% belong to non-white minority groups. The education level is relatively high, with 64.2% having attended university.

Pre-treatment income is mainly concentrated in the middle and lower-middle brackets (22.9% in 15k–24k and 24.2% in 25k–39k), while 18.2% fall below 15k, indicating high vulnerability. The average baseline level of depressive symptoms (`depress_pre`) is 1.87.

Regarding the intervention, 66.7% of participants were assigned to the training programme ($Z=1$), but only 41.4% actually participated ($W=1$), revealing substantial non-compliance.

4.2 Subset Analysis by Assignment

Comparing control ($Z=0$) and treatment ($Z=1$) groups shows excellent covariate balance. The mean age is 37.3 in the control group and 37.7 in the treatment group, while the proportion of individuals with university education is 63.9% and 64.3%, respectively. Pre-treatment depressive symptoms are also very similar (1.89 vs 1.86), and income distributions are nearly identical across groups.

Importantly, the mean of W in the control group is exactly 0, confirming one-sided non-compliance: individuals not assigned to treatment could not access the programme. Among those assigned ($Z=1$), 62% complied (compilers) and 38% did not participate (never-takers).

4.3 Subset Analysis by Receipt Indicators

In this case the situation radically changes, in fact when we apply the stratification based on the behavior of the units, by comparing those who didn't participate and who did. In this case we can see a significant imbalance in covariates:

- **Age:** Who participate at the program are significantly older.
- **Sex, non white:** Among the participants, there is a lower proportion of both women and non-white individuals.
- **educ, income:** Among the participants, there is a higher proportion of people who have a higher education and are in the highest income bracket.

In the figure below we can see the image that shows the percentage of the mean in the different dataset.

[immagine plot delle medie]

5 Outcome Analysis

In this section, we analyze how the treatment influences the primary outcome (`job_seek`), which measures the intensity and psychological effort put into looking for work.

- **Full Dataset (Left Panel):** The near-zero density at scores 1 and 2 reveals the absence of completely passive or deeply discouraged individuals. The sharp surge starting at 3 shows that the state of unemployment itself triggered an immediate mobilization response, with most subjects highly active even before the intervention.
- **Assignment Effect :** The randomized invitation acted as a clear psychological trigger, the assigned group ($Z = 1$, purple) shows lower density around medium scores (3–3.5) but overtakes the control group ($Z = 0$, yellow) near the upper bound, creating a second peak at the maximum score (5). This visually captures the clean.
- **Participation Comparison:** Non-participants ($W = 0$, red) are concentrated at lower-medium levels (2–3), while actual participants ($W = 1$, blue) peak near 4 and 5, with virtually no presence in the lower ranges, while this suggests a massive treatment effect, this gap is inherently confounded by self-selection: the blue line reflects both the actual benefit of the course and the fact that participants possessed stronger baseline characteristics (older, wealthier, and more motivated) from the start.

6 Analysis of Unconfoundedness

In this section we analyze the **unconfoundedness** in the assignment and in the treatment receipt.

6.1 Unconfoundedness of Assignment

we can assume that the assignment to treatment is unconfounded because:

1. **Theoretical Motivation:** We know the JobsII study is controlled randomized experiment, so the assignment mechanism is known and not confounding, this ensures ex ante that Z is independent of any characteristics of the subjects, whether observable or unobservable.
2. **Empirical evidence from data:** Looking at the means stratified by Z , the pre-treatment covariates are perfectly balanced. This homogeneity confirms the success of the randomisation.

6.2 Unconfoundedness of Treatment Receipt

In this case we can't assume that the effective receipt are **unconfoundedness** of treatment W .

1. **Theoretical Motivation:** Unlike Z , the variable W is the result of the participants' free choice (phenomenon known as non-compliance). Even after controlling for observed covariates, we cannot rule out the presence of unobserved confounders like family support or other, which simultaneously influence the decision.
2. **Empirical evidence from data:** Looking at the means stratified by W , a strong systematic imbalance emerges at the outset, in fact we can notice who choose to participate compared with those who do not are significantly older 39.2 vs 36.4 years, have a lower proportion of women 47.8% vs 57.7% and of minorities 15% vs 18.2%. Another important imbalance is highlighted by the income in fact have more individuals in the highest income bracket.

6.3 The importance of Unconfoundedness

The distinction of Assigned treatment and treatment receipt is central for causal identification, in fact a direct comparison between treated units and control units in general don't identify a causal effect, because the participation is not random. The units that decide to take part may differ from others by observable and not observable characteristics, such as, motivation, personal condition or family support. Consequently the quantity

$$\mathbb{E}[Y|W = 1] - \mathbb{E}[Y|W = 0] \quad (1)$$

Doesn't match with true causal effect of treatment $\mathbb{E}[Y(1) - Y(0)]$, since is contaminated by selection bias. To resolve this issue, we use the causal variable for the Z assignment, generated by randomization, in fact the Z variable is independent from potential outcome by satisfying:

$$Z \perp (Y(0), Y(1), W(0), W(1)) \quad (2)$$

This guarantees that Z is a source of exogenous variation, that is, a variable determined outside the model and therefore not influenced by unobserved factors that also affect the outcome. The significance of this result lies in the fact that it allows us to distinguish between correlation and causation: whilst W is endogenous due to individual selection, Z represents a variation external to individuals' behaviour. For this reason, Z enables us to analyse the programme's effect reliably even in the presence of non-compliance, whilst maintaining a valid causal interpretation.

7 Intention-To-Treat Effect

The **Intention-to-Treat (ITT)** effect into primary outcome measures the causal impact that the program might have in research intensity for the units work, comparing the outcome mean between treated and control group, respectively $Z=1$, $Z=0$. Since it is an estimate based on originally randomization, the ITT

is mathematically protected by the bias of auto-selection and quantify the real efficiency of intervention policy. Randomization allows for the identification and estimation of the average ITT effect, and the formula for the primary outcome is:

$$\tau_Y = \mathbb{E}[Y_i^{obs}|Z_i = 1] - \mathbb{E}[Y_i^{obs}|Z_i = 0] \quad (3)$$

Under CRE we can estimate τ_Y follow the Neyman approach using the simple difference-in-means estimators because the Neyman's estimator of the variance can be used for estimate the sampling uncertainty of these estimators.

$$\hat{\tau}_Y = \bar{Y}_1^{obs} - \bar{Y}_0^{obs} \quad (4)$$

In the JobsII study the ITT are $\hat{\tau}_Y = 0.06745$, this means that the program reach the main goal, since send the invite to participate at the program pushed the units to search job with more intensity, discuss of result based on causal assignment this increase is a certainly causal effect, that mean it don't depend by other units features.

8 Compliance Type

Compliance status refers to a unit's response to the assignment, for both values of the assignment whether that status is observed or unobserved Formally, the compliance type is a function of the pair of potential responses:

$$Gi = (W_i(0), W_i(1)) \quad (5)$$

Because both $W_i(0)$ and $W_i(1)$ are binary indicators, there are four possible values for the pair of potential responses to treatment assignment:

TO DOOOOO con tabella quella che mette anche lei

- **Compliers:** Units who always comply with their assignment in the context of this study
- **Never-takers:** Units who never take the treatment, irrespective of their assignment in this study
- **Always-takers:** Units who always take the treatment, irrespective of their assignment in this study
- **Defiers:** Units who, in the context of this study, irrespective of the value of their assignment, would do the opposite of their assignment: they are induced to avoid the treatment by assignment to it

In the context of this study, since we are in one-side non-compliance, the **Always-taken** and the **Defiers** don't exist by design because only those units assigned to the active treatment could choose whether or not to comply with their assignment.

9 Monotonicity Assumption

The **monotonicity assumption** that being assigned to the treatment group ($Z=1$) can increase or leave the probability to actually take the treatment (W), but don't decrease it compared to being assigned to control group ($Z=0$). The monotonicity assumption has a testable implication that

$$\Pr(W_i^{obs} = 1|Z = 1) \geq \Pr(W_i^{obs} = 1|Z = 0) \quad (6)$$

This assumption holds out the **Defiers**. In this case of study the monotonicity assumption is satisfied automatically, because for one-side non compliance the Defiers do not exist, and the equation is already satisfied.

10 Exclusion Restriction

10.1 Exclusion Restriction Never-takers

The **exclusion restriction** established that the causal assignment at the treatment affect the outcome exclusively through the effective participation at the treatment. In the specific case of **Never-takers**, the assumption required that the our potential outcome are identical regardless of the group to which they have been assigned.

$$Y_i(Z = 1, W = 0) = Y_i(Z = 0, W = 0) \quad (7)$$

In the study case the letter of invitation don't have impact on the individual's well-being.

10.1.1 Ground suspicion

In the context of the JobsII, the potential outcome, `job_seek`, measures an individual's sense of security and motivation when facing job interviews, so the reason for scepticism might be:

An unemployed person who feels discouraged receives an official letter informing them that they have been selected in to program, even if they decide to not to attend the session, the mere fact of knowing that an institution has demanded them an ideal candidate and "worthy" of attention can act as a confidence boost. This validation can boost their self-esteem and their belief that they can succeed, leading them to achieve a higher `job_seek` score than they would have had if they don't placed in the control group without receiving invitation.

10.1.2 Plausibility Assessment

This concern should be regarded as irrelevant. The `job_seek` variable doesn't measure a temporary emotion, but rather the perception of practical and complex skills, therefore a simple formal notification lacks the power to transmit

these cognitive skills, which are developed exclusively through active participation and piratical interactions during the seminarious. It's possible to conclude that the entirely valid and sensible to maintain the **Exclusion Restriction** assumption for **Never-Takers**.

10.2 Exclusion Restriction Always-taker

Given that the JobssII study features a one-sided noncompliance design, the **Always-taker** category is not present, consequentially the issue of a breach of the exclusion restriction doesn't arise for this specific group.

11 Complier Average Causal Effect (CACE)

The **CACE** measure the mean causal effect of the treatment on **Compliers**, i.e. those who follow the random assignment, for estimate this we use the **Wald Estimator**, that compares the effect of random allocation on the primary outcome with the effect of allocation on actual participation in the treatment. The formula of Wald estimator or IV estimator:

$$\hat{\tau}_{Y,C} = \frac{\hat{\tau}_Y}{\hat{\tau}_W} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{W}_1 - \bar{W}_0} \quad (8)$$

The result in the Jobs study are:

$$\hat{\tau}_{Y,C} = 0.10879 \quad (9)$$

The estimated CACE indicates that the treatment has a positive causal effect on job search motivation and preparatory effort for interviews among compliers, i.e. individuals whose participation is induced by assignment. To assess the accuracy of the Wald estimator and express its sampling uncertainty, we calculate its asymptotic standard error using the Delta Method.

$$\hat{V}(\hat{\tau}_{Y,C}) = \frac{1}{\hat{\tau}_W^2} \hat{V}(\hat{\tau}_Y) + \frac{\hat{\tau}_Y^2}{\hat{\tau}_W^4} \hat{V}(\hat{\tau}_W) - 2 \frac{\hat{\tau}_Y}{\hat{\tau}_W^3} \hat{C}(\hat{\tau}_W, \hat{\tau}_Y) \quad (10)$$

where the standard error is:

$$\hat{se} = \sqrt{\hat{V}(\hat{\tau}_{Y,C})} \quad (11)$$

The CACE estimate of 0.109 indicates that the JOBS II programme has a positive impact on self-efficacy in job-seeking (job_seek) amongst 'Compliers' – that is, those who participate in the course only if formally invited. To assess the precision of this estimate, we calculated the standard error using the Delta

Method $\hat{se} = 0.081$ and the corresponding 95% confidence interval $(1 - \alpha)$ which is $[-0.050, 0.268]$ that in this context the estimate implies, although the effect on participants is quantitatively positive, the estimate statistically imprecise, as the confidence interval is wide and includes zero, we do not have sufficient statistical power to reject the null hypothesis. Consequently, we conclude that the positive impact observed in the sample could be due solely to sampling variability and not to a real and robust structural change in the population of **Compliers**

12 CACE vs ITT

The CACE estimate compared with the overall ITT (Intention-To-treatment) estimate of the outcomes, assessing its informative value from two different decision-making perspectives. In a mathematical point, given the absence of **Always-Takers**, as seen, the relationship between two estimators is defined by the formula below:

$$ITT_Y = CACE \times \pi_C \quad (12)$$

where π_C is the proportion (quota) of **Compliers** in the population. Consequently, the ITT_Y will be numerically lower than the CRE. We have two types of perspective policy:

- **Public policy perspective :** **ITT** estimate is far more informative and valuable, those designing a public intervention can only control the allocation of the treatment, but cannot physically force individuals to participate; therefore, the **ITT** captures the programme's actual impact on the entire target population, already incorporating factoring in individual refusals.
- **Individual policy perspective:** who must decide whether or not to take part in the program, the **CACE** estimator is far more relevant; the unemployed person is not interested in the average effect on those who reject the program effectiveness, they want to know the actual impact on their job-seeking behavior if they decide to join the invitation. The **CACE** answers the simple question: "if I choose to actively participate in the course, what concrete benefit will I derive from it?"

In conclusion, in the context of JobsII the CACE shows that for the individual participant the estimated effect is positive, although statistically imprecise, suggesting that it is still the most relevant estimator as it demonstrates the direct effects of those who choose to take part in treatment.

13 Method of Moment

13.1 Theoretical Introduction

In this section we estimate a treatment model with non-compliance using the **method of moments**, in which the structural parameters are identified by impos-

ing conditions on the theoretical and sample moments, the estimate are obtained via 50.000 bootstrap replicates, in order to assess the numerical stability and statistical uncertainty of the parameters, including group proportion, conditional means and causal effect. The **Moments Method** is used for estimate the parameters imposing that the theoretical values of the model coincide with those observed in the sample, allow the identification even in the presence of latent variable like compliance type. The **Bootstrap** is used for approximate the sample distributions of estimator and obtain the standard error with confidence interval without making any assumptions about their asymptotic behavior. In the Jobs study case we have the one-side non-compliance, where the Always taken is not defined, and this change how to formulate the calculus for the moment:

$$\hat{\pi}_{AT} = \overline{W}_0^{obs} = 0 \quad (13)$$

$$\hat{\pi}_{NT} = 1 - \overline{W}_1^{obs} \quad (14)$$

$$\hat{\pi}_C = \overline{W}_1^{obs} \quad (15)$$

This involves:

$$\hat{\mu}_{0C} = \left(\overline{Y}_{z=0,w=0} - \frac{\hat{\pi}_{NT}}{\hat{\pi}_C + \hat{\pi}_{NT}} \overline{Y}_{z=1,w=0} \right) \frac{\hat{\pi}_C + \hat{\pi}_{NT}}{\hat{\pi}_C} \quad (16)$$

$$\hat{\mu}_{1C} = \left(\overline{Y}_{z=1,w=1} - \frac{\hat{\pi}_{AT}}{\hat{\pi}_C + \hat{\pi}_{AT}} \overline{Y}_{z=0,w=1} \right) \frac{\hat{\pi}_C + \hat{\pi}_{AT}}{\hat{\pi}_C} \quad (17)$$

So let's see how we've simplified the formula 16):

$$\begin{aligned} \hat{\pi}_{AT} &= \overline{W}_0^{obs} = 0 \\ \hat{\pi}_{NT} &= 1 - \overline{W}_1^{obs} \\ \hat{\pi}_C &= \overline{W}_1^{obs} - \overline{W}_0^{obs} \\ \text{Combining the formula} \\ \hat{\pi}_C + \hat{\pi}_{NT} &= 1 \\ \hat{\mu}_{0C} &= \frac{(\overline{Y}_{z=0,w=0} - \hat{\pi}_{NT} \overline{Y}_{z=1,w=0})}{\hat{\pi}_C} \end{aligned} \quad (18)$$

So let's see how we've simplified the formula 17):

$$\begin{aligned} \hat{\pi}_{AT} &= \overline{W}_0^{obs} = 0 \\ \hat{\pi}_{NT} &= 1 - \overline{W}_1^{obs} \\ \hat{\pi}_C &= \overline{W}_1^{obs} - \overline{W}_0^{obs} \\ \text{Combining the formula} \\ \hat{\mu}_{1C} &= \left(\overline{Y}_{z=1,w=1} - \frac{\overset{=0}{\hat{\pi}_{AT}}}{\cancel{\hat{\pi}_C} + \hat{\pi}_{AT}} \overline{Y}_{z=0,w=1} \right) \frac{\overset{=1}{\hat{\pi}_C} + \cancel{\hat{\pi}_{AT}}}{\hat{\pi}_C} \\ \hat{\mu}_{1C} &= \overline{Y}_{z=1,w=1} \end{aligned} \quad (19)$$

13.2 Estimation Results

When applying the Method of Moments estimators to the JOBS II dataset for the outcome of self-efficacy in job seeking (job_seek), the following point estimates are obtained. Remember the type of compliance is a latent, invariant characteristic and, due to randomisation, has the same distribution in both arms; π_c is identified in arm Z=1 via W, whilst in arm Z=0, compliers and never-takers are indistinguishable (W=0).

For read the table below follow this legend.

Symbol	Description
$\hat{\pi}_C$	Complier ratio
$\hat{\pi}_{NT}$	Never-takers ratio
$\bar{Y}_{NT}$	Outcome mean for never-takers
$\bar{Y}_C(1)$	Outcome mean for compliers under treatment
$\bar{Y}_C(0)$	Outcome mean for compliers under control
CACE	Complier Average Causal Effect

Table 3: Notation legend

These, on the other hand, are the results after applying bootstrapping 2,000 and 50,000 times to calculate the standard error as well, in the first tabel is reported the 2000 times and in the second 50000

Table 4: Application of Moments Methods on JobsII

Parameter	Point Estimate (MoM)	$\hat{SE}$	CI
$\hat{\pi}_C$	0.6200	0.0198	[0.5812 , 0.6588]
$\hat{\pi}_{NT}$	0.3800	0.0198	[0.3412 , 0.4188]
$\bar{Y}_{NT}$	3.9861	0.0549	[3.8785 , 4.0937]
$\bar{Y}_c(1)$	4.1146	0.0352	[4.0456 , 4.1836]
$\hat{Y}_C(0)$	4.0058	0.0745	[3.8598 , 4.1518]
CACE	0.1088	0.0824	[-0.0527 , 0.2703]

Table 5: Application of Moments Methods on JobsII

Parameter	Point Estimate (MoM)	$\hat{SE}$	CI
$\hat{\pi}_C$	0.6200	0.0199	[0.5810 , 0.6590]
$\hat{\pi}_{NT}$	0.3800	0.0199	[0.3410 , 0.4190]
$\bar{Y}_{NT}$	3.9861	0.0552	[3.8779 , 4.0943]
$\bar{Y}_c(1)$	4.1146	0.0352	[4.0456 , 4.1836]
$\hat{Y}_C(0)$	4.0058	0.0731	[3.8625 , 4.1491]
CACE	0.1088	0.0811	[-0.0502 , 0.2678]

Analysis of the JOBS II programme reveals that the sample comprises 62% ‘Compliers’ and 38% ‘Never-takers’. In terms of the actual impact on self-efficacy in job-seeking, the course has a positive effect on actual participants, raising the average score of the ‘Compliers’ from 4.0058 in the control group to 4.1146 in the treatment group, with a CACE of 0.1088. By contrast, the overall effect of assignment (ITT) stands at 0.0675, as the presence of Never-takers dilutes the effectiveness of the intervention on a population-wide scale. However, the standard error of the CACE calculated via bootstrapping is 0.0811, yielding a 95% confidence interval ranging from -0.0502 to 0.2678. as this interval includes zero, the effect of the course does not reach statistical significance, highlighting how the Method of Moments tends to amplify sampling variability when mathematically reconstructing latent data.